

Design and Hedging of a Bonus Certificate

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1. Introduction and Original Data Import

In this report, we design and value a Bonus Certificate (BC) on an assigned equity using market data sourced from Yahoo Finance. The BC is composed into a long position in the underlying share and a down-and-out barrier put, which replicates the structured payoff while enabling the issuing bank to hedge risk and still has margin. The valuation framework employs the Bates stochastic volatility model, in which we implement the Carr–Madan FFT algorithm to price vanilla calls and puts and calibrate model parameters to observed implied volatilities, then apply Monte Carlo simulation to compute the exotic barrier-in option component and obtain the BC's theoretical fair value. Finally, we derive the BC's delta under the calibrated Bates dynamics and propose a delta-hedging strategy.

2. Original Data Import

Market and option price data for both the underlying equity and its vanilla calls and puts were sourced directly from *Yahoo Finance*¹. All analyses are conducted from the pricing date May 16, 2025 (which we set $t = 0$) with the structured product set maturing on September 19, 2025, and the full set of call and put quotes across strikes along with their implied volatilities for that expiry are shown below (Figures 1 & 2 & 4 & 5). To incorporate risk-free rates into the analysis, the US Treasury interest rate was used as an equivalent². The risk-free interest rate term structure was taken from World Government Bonds data, as illustrated in Figure 3.

NYSE - Nasdaq Real Time Price • USD

Delta Air Lines, Inc. (DAL) ☆ Follow + Add holdings Time to buy DAL?

50.46 -0.12 (-0.24%)
As of 1:15:30 PM EDT. Market Open.

Sep 19, 2025 ▾ All Strike Prices ▾ List ▾ Calls ▾

Calls In The Money

Contract Name	Last Trade Date (EDT)	Strike	Last Price	Bid	Ask	Change	% Change	Volume	Open Interest	Implied Volatility
DAL250919C00020000	4/28/2025 9:46 AM	20	23.15	30.55	30.90	0.00	0.00%	2	28	87.70%
DAL250919C00023000	5/12/2025 3:15 PM	23	28.32	27.65	28.20	0.00	0.00%	13	26	85.55%
DAL250919C00024000	4/7/2025 2:40 PM	24	15.25	0.00	0.00	0.00	0.00%	-	0	0.00%
DAL250919C00025000	5/9/2025 1:22 PM	25	23.30	25.75	26.00	0.00	0.00%	1	9	75.59%
DAL250919C00026000	4/8/2025 1:36 PM	26	12.85	22.75	23.20	0.00	0.00%	11	12	0.00%
DAL250919C00027000	4/4/2025 12:04 PM	27	12.20	17.90	18.15	0.00	0.00%	3	3	0.00%
DAL250919C00028000	4/7/2025 2:17 PM	28	12.10	0.00	0.00	0.00	0.00%	2	0	0.00%
DAL250919C00029000	5/9/2025 9:39 AM	29	20.70	22.00	22.20	0.00	0.00%	1	6	68.80%
DAL250919C00030000	5/6/2025 3:42 PM	30	20.66	21.05	21.20	4.71	29.53%	2	28	66.02%

Figure 1 Market Price of Call Option

¹ <https://finance.yahoo.com/quote/DAL/options/?date=1758240000&type=puts>

² https://www.worldgovernmentbonds.com/country/united-states/?utm_content=cmp-true

DAL250919C00031000	5/13/2025 12:32 PM	31	20.15	20.15	20.45	-1.90	-8.62%	2	5	66.89%
DAL250919C00032000	5/12/2025 3:29 PM	32	19.92	19.20	19.45	0.00	0.00%	22	22	64.01%
DAL250919C00033000	5/12/2025 12:07 PM	33	18.42	18.25	18.50	0.00	0.00%	1	300	61.67%
DAL250919C00034000	5/8/2025 12:16 PM	34	15.65	17.40	17.60	0.00	0.00%	2	23	60.74%
DAL250919C00035000	5/8/2025 12:16 PM	35	14.80	16.50	16.75	0.00	0.00%	5	148	59.62%
DAL250919C00036000	5/5/2025 10:32 AM	36	11.80	15.55	16.10	0.00	0.00%	1	128	59.52%
DAL250919C00037000	5/9/2025 1:22 PM	37	12.65	14.85	15.10	0.00	0.00%	2	162	58.37%
DAL250919C00038000	5/15/2025 11:33 AM	38	13.95	14.00	14.10	-2.25	-13.89%	2	216	55.98%
DAL250919C00039000	5/15/2025 9:47 AM	39	13.10	13.15	13.30	-0.94	-6.70%	1	100	54.91%
DAL250919C00040000	5/14/2025 2:40 PM	40	12.35	12.35	12.50	-0.30	-2.37%	3	527	53.96%
DAL250919C00041000	5/15/2025 9:31 AM	41	11.80	11.60	11.75	-1.25	-9.58%	8	266	53.39%
DAL250919C00042000	5/14/2025 2:34 PM	42	11.15	10.85	10.95	0.00	0.00%	104	770	52.32%
DAL250919C00043000	5/14/2025 2:08 PM	43	10.64	10.05	10.20	0.00	0.00%	7	205	51.05%
DAL250919C00044000	5/14/2025 2:34 PM	44	9.70	9.40	9.55	0.00	0.00%	1	405	50.88%
DAL250919C00045000	5/15/2025 10:31 AM	45	8.25	8.70	8.85	-0.98	-10.62%	100	3,212	50.71%
DAL250919C00046000	5/15/2025 10:38 AM	46	7.55	8.05	8.20	-1.85	-19.68%	1	484	50.02%
DAL250919C00047000	5/15/2025 12:14 PM	47	7.49	7.45	7.55	-1.56	-17.24%	2	1,388	49.15%
DAL250919C00048000	5/15/2025 10:31 AM	48	6.45	6.80	6.95	-0.70	-9.79%	100	440	48.49%
DAL250919C00049000	5/14/2025 12:10 PM	49	7.27	6.15	6.40	0.00	0.00%	15	2,905	48.04%
DAL250919C00050000	5/15/2025 12:07 PM	50	5.70	5.70	5.80	-0.35	-5.79%	117	3,040	46.94%
DAL250919C00052500	5/15/2025 12:28 PM	52.5	4.49	4.50	4.65	-0.26	-5.47%	36	1,741	46.18%
DAL250919C00055000	5/15/2025 11:44 AM	55	3.55	3.50	3.60	-0.25	-6.58%	6,110	7,661	44.85%
DAL250919C00057500	5/15/2025 11:54 AM	57.5	2.67	2.70	2.76	-0.20	-6.97%	166	3,309	43.92%
DAL250919C00060000	5/14/2025 3:10 PM	60	1.91	1.97	2.09	-0.26	-11.98%	1	5,933	43.21%
DAL250919C00062500	5/15/2025 12:09 PM	62.5	1.55	1.52	1.57	-0.13	-7.74%	50	719	42.69%
DAL250919C00065000	5/14/2025 3:57 PM	65	1.14	1.13	1.18	-0.07	-5.79%	11	7,026	42.43%
DAL250919C00067500	5/15/2025 9:43 AM	67.5	0.84	0.84	0.93	-0.16	-16.00%	3	1,374	42.92%
DAL250919C00070000	5/15/2025 11:14 AM	70	0.62	0.61	0.65	-0.06	-8.82%	15	1,624	41.99%
DAL250919C00072500	5/12/2025 2:54 PM	72.5	0.40	0.46	0.49	0.00	0.00%	1	631	42.09%
DAL250919C00075000	5/14/2025 10:03 AM	75	0.47	0.34	0.37	0.00	0.00%	2	2,056	42.24%
DAL250919C00077500	5/15/2025 10:59 AM	77.5	0.26	0.26	0.39	+0.01	+4.00%	2	113	45.31%
DAL250919C00080000	5/13/2025 3:32 PM	80	0.26	0.06	0.36	0.00	0.00%	1	200	47.02%

Figure 2 Market Data of Call Option (2)

	Residual Maturity	Annualized Yield				Constant Maturity Zero Coupon Price				Capital Growth of 1 Unit	Last Update
		Last	Chg 1M	Chg 6M	Chg 12M	Last	Chg 1M	Chg 6M	Chg 12M		
	1 month	4.298%	-0.4 bp	-29.6 bp	-106.2 bp	99.65	0.00 %	+0.02 %	+0.08 %	1.003	16 May
	2 months	4.315%	+1.2 bp	-24.4 bp	-105.8 bp	99.30	0.00 %	+0.04 %	+0.17 %	1.007	16 May
	3 months	4.354%	+3.7 bp	-15.8 bp	-104.9 bp	98.94	-0.01 %	+0.04 %	+0.25 %	1.010	16 May
	4 months	4.355%	+1.7 bp	-16.7 bp	-104.5 bp	98.59	0.00 %	+0.05 %	+0.34 %	1.014	16 May
	6 months	4.269%	+8.0 bp	-21.7 bp	-109.7 bp	97.93	-0.04 %	+0.10 %	+0.52 %	1.021	16 May
	1 year	4.130%	+17.1 bp	-21.1 bp	-100.0 bp	96.03	-0.17 %	+0.20 %	+0.96 %	1.041	16 May

Figure 3 World Government Bonds Interest Rate

50.45 -0.13 (-0.26%)

As of 1:17:07 PM EDT. Market Open.

Sep 19, 2025

All Strike Prices

List

Puts

Puts

In The Money

Contract Name	Last Trade Date (EDT)	Strike	Last Price	Bid	Ask	Change	% Change	Volume	Open Interest	Implied Volatility
DAL250919P00020000	5/15/2025 10:59 AM	20	0.11	0.03	0.25	+0.02	+22.22%	3	44	79.88%
DAL250919P00021000	4/21/2025 1:32 PM	21	0.55	0.03	0.26	0.00	0.00%	8	10	76.47%
DAL250919P00022000	5/12/2025 9:45 AM	22	0.12	0.01	0.29	0.00	0.00%	1	5	73.24%
DAL250919P00023000	5/5/2025 12:30 PM	23	0.28	0.01	0.31	0.00	0.00%	2	70	70.31%
DAL250919P00024000	5/8/2025 12:33 PM	24	0.24	0.06	0.33	0.00	0.00%	1	1	69.34%
DAL250919P00025000	5/12/2025 3:41 PM	25	0.23	0.00	0.36	0.00	0.00%	1	121	65.04%
DAL250919P00026000	5/1/2025 9:40 AM	26	0.64	0.01	0.39	0.00	0.00%	8	29	62.99%
DAL250919P00027000	5/5/2025 1:41 PM	27	0.48	0.04	0.42	0.00	0.00%	40	117	61.43%
DAL250919P00028000	5/8/2025 3:18 PM	28	0.39	0.20	0.46	0.00	0.00%	91	127	62.70%
DAL250919P00029000	5/12/2025 3:40 PM	29	0.30	0.11	0.51	0.00	0.00%	3	18	58.69%
DAL250919P00030000	5/13/2025 3:40 PM	30	0.30	0.25	0.55	0.00	0.00%	12	10,661	58.89%
DAL250919P00031000	5/14/2025 2:12 PM	31	0.35	0.39	0.42	0.00	0.00%	8	2,804	55.96%
DAL250919P00032000	5/9/2025 9:34 AM	32	0.67	0.00	0.49	0.00	0.00%	17	59	55.42%
DAL250919P00033000	5/14/2025 3:24 PM	33	0.53	0.53	0.56	0.00	0.00%	9	4,708	53.81%
DAL250919P00034000	5/12/2025 3:41 PM	34	0.55	0.60	0.69	0.00	0.00%	8	193	53.13%
DAL250919P00035000	5/14/2025 3:51 PM	35	0.73	0.71	0.77	0.00	0.00%	5	787	52.10%
DAL250919P00036000	5/15/2025 11:45 AM	36	0.81	0.82	0.88	+0.04	+5.19%	1	495	51.12%
DAL250919P00037000	5/12/2025 11:47 AM	37	0.93	0.95	1.12	0.00	0.00%	2	3,579	51.22%
DAL250919P00038000	5/14/2025 3:06 PM	38	1.08	0.94	1.22	0.00	0.00%	5	2,497	50.95%
DAL250919P00039000	5/13/2025 1:50 PM	39	1.30	1.23	1.32	+0.34	+35.42%	11	290	49.15%
DAL250919P00040000	5/15/2025 11:50 AM	40	1.50	1.29	1.48	+0.02	+1.35%	9	10,166	48.05%
DAL250919P00041000	5/14/2025 2:17 PM	41	1.60	1.65	1.88	0.00	0.00%	32	472	49.71%
DAL250919P00042000	5/14/2025 3:00 PM	42	1.87	1.87	1.94	0.00	0.00%	173	2,535	46.95%
DAL250919P00043000	5/15/2025 12:09 PM	43	2.18	2.13	2.18	+0.05	+2.35%	8	284	46.14%
DAL250919P00044000	5/14/2025 3:47 PM	44	2.43	2.41	2.53	0.00	0.00%	187	437	46.22%
DAL250919P00045000	5/14/2025 3:07 PM	45	2.93	2.72	2.78	+0.24	+8.92%	2	2,127	45.02%

Figure 4 Market Data of Put Option (1)

DAL250919P00046000	5/14/2025 3:14 PM	46	3.00	3.00	3.15	0.00	0.00%	306	308	44.75%
DAL250919P00047000	5/14/2025 3:21 PM	47	3.40	3.40	3.50	0.00	0.00%	122	2,543	44.02%
DAL250919P00048000	5/14/2025 2:49 PM	48	3.75	3.80	3.90	0.00	0.00%	23	110	43.51%
DAL250919P00049000	5/14/2025 1:45 PM	49	3.92	4.25	4.35	0.00	0.00%	7	65	43.16%
DAL250919P00050000	5/15/2025 10:33 AM	50	4.95	4.70	4.80	+0.27	+5.77%	8	2,777	42.58%
DAL250919P00052500	5/15/2025 11:08 AM	52.5	6.05	5.90	6.05	+0.95	+18.63%	6	1,283	41.04%
DAL250919P00055000	5/14/2025 3:45 PM	55	7.35	7.40	7.55	0.00	0.00%	21	1,178	40.02%
DAL250919P00057500	5/15/2025 9:53 AM	57.5	9.35	9.00	9.20	+1.55	+19.87%	10	824	38.70%
DAL250919P00060000	5/12/2025 11:36 AM	60	10.60	10.90	11.05	0.00	0.00%	1	3,858	37.65%
DAL250919P00062500	5/14/2025 2:47 PM	62.5	12.78	12.90	13.05	0.00	0.00%	1	669	36.57%
DAL250919P00065000	5/8/2025 11:52 AM	65	17.25	15.05	15.15	0.00	0.00%	8	487	34.99%
DAL250919P00067500	4/9/2025 3:42 PM	67.5	23.30	19.30	19.55	0.00	0.00%	4	17	60.18%
DAL250919P00070000	4/3/2025 3:19 PM	70	31.30	25.70	26.05	0.00	0.00%	250	3	100.79%
DAL250919P00072500	3/13/2025 10:18 AM	72.5	27.18	30.10	33.45	0.00	0.00%	1	1	133.89%
DAL250919P00075000	2/4/2025 3:09 PM	75	10.43	21.00	22.25	0.00	0.00%	2,002	2,684	0.00%
DAL250919P00077500	3/4/2025 10:11 AM	77.5	23.90	0.00	0.00	0.00	0.00%	-	0	0.00%

Figure 5 Market Data of Put Option (2)

3. Valuation of the Bonus Certificate (BC)

3.1. Decomposition into Building Blocks

A Bonus Certificate can be replicated synthetically using the following building blocks:

Long a stock at the beginning:

Hold a stock to capture upside above B and deliver S_T when the barrier is breached or $S_T \geq B$.

Down-and-Out Barrier Put (DO Put):

To replicate the “bonus” payoff of the Bonus Certificate, the bank takes a long position in a down-and-out barrier put with strike $K = B$ and barrier H . If the stock never falls below H and finishes at or below B , the option pays exactly the shortfall $(B - S_T)$ and if the barrier is breached at any time, or if $S_T > B$, the put “knocks out” and pays nothing. As sum, it can be described as follows:

$$\begin{cases} B - S_T, & \text{if } H < S_t \leq B \forall t \leq T \\ 0, & \text{otherwise} \end{cases}$$

Risk-free account:

The issuer can put extra money into the risk-free bank account that grows at risk free rate at maturity T

In summary, the payoff is:

$$\text{Bonus Certificate} = S_T + (B - S_T)_{\text{knock-out if } S_t \leq H}^+$$

Which recovers the three scenarios:

- $S_T > B$: investor receives S_T
- $H < S_T \leq B$, $\min S_t > H$: investor receives B
- Barrier touched: investor receives S_T

The cash flow at the moment t can be summarized as Table 1 Cashflow at the moment tTable 1:

Building block	Long / Short	Cash-flow	Description
Cash	Short	V_{BC}	Cash get from selling BC

Stock	Long	$-S_t$	Provides unlimited upside participation
Down-and-Out Barrier Put($K = B, H$)	Long	$-P_{KO}$	Generates the “bonus” if the barrier is never breached
Risk-free bank account	Short	$-(V_{BC} - S_t - P_{KO})$	Ensures net cost = 0 today while leaving a positive spread for the bank

Table 1 Cashflow at the moment t

The cashflow at maturity is summarized as Table 2:

Scenarios	Short 1 stock	DO- Put	Payoff of BC	Cash withdraw from risk-free bank account	Net cashflow
$S_T > B$	$+S_t$	0	$-S_t$	$+(V_{BC} - S_t - P_{KO})e^{rT}$	$+(V_{BC} - S_t - P_{KO})e^{rT}$
$H \leq S_T < B$	$+S_t$	$+B - S_T$	$-B$	$+(V_{BC} - S_t - P_{KO})e^{rT}$	$+(V_{BC} - S_t - P_{KO})e^{rT}$
$S_T < B$	$+S_t$	0	$-S_t$	$+(V_{BC} - S_t - P_{KO})e^{rT}$	$+(V_{BC} - S_t - P_{KO})e^{rT}$

Table 2 Cashflow at maturity

In both the “barrier hit” and “upside” scenarios, the barrier put expires worthless, leaving only to deliver S_T , matching the BC payoff of S_T .

In the BC’s middle scenario (“barrier intact” and $S_T \leq B$), the investor is entitled to a fixed bonus B . This can be done by combining this barrier put payout ($B - S_T$) with delivering S_T .

In all three scenarios, there is always a cashflow in, as long as the price of BC is larger than the sum of Down-and-Out Barrier put option and stock price at the start, which is also the cost of the bank.

3.2. Why investors like it?

If the stock trades between 80% and 120% of today’s level, investors lock in a 10% gain at maturity, significantly outperforming a risk-free deposit. Furthermore, the 20% barrier buffer reduces the chance of receiving only the stock value, shielding investors from modest corrections. Should the stock rally above 120%, investors fully participate in further gains, unlike typical capped-return structures.

3.3. Code Overview

To compute the price of the Bonus Certificate (BC) under the Bates model, we implemented three modular algorithms in MATLAB. First, the parameters must be set. To balance investor appeal and bank’s margin, we select variables summarized in Table 3:

Symbol	Value	Short justification
Maturity T	0,345	From 05-16-2025 to 09-19-2025(126/365)
Stock Price	50,46	The spot observed at 05-16
Bonus level B	60,55	+20 % above spot, which offers an attractive capped upside if markets remain flat or moderately bullish.
Barrier H	40,37	−20 % below spot which absorbs moderate dips but leaves investors exposed only to severe drawdowns.
Dividend yield q	1,32%	Latest forward dividend yield
Risk-free rate r	4,355 %	4-month UST constant-maturity zero-rate

Table 3 Overview of variables

4. Valuation and Delta Hedging

4.1. Vanilla Call/Put Pricer via FFT & Carr–Madan

We implement a fast Carr–Madan FFT pricer for vanilla calls and puts by computing the discounted risk-neutral expectation via the characteristic function of the Bates process. In the code, the Bates model’s characteristic function is defined as $\varphi(u)$, by combining the Heston stochastic-volatility dynamics with a Merton jump component and including the necessary jump–drift compensation. This function accepts a complex frequency vector and outputs $\varphi(u)$ for use in Fourier pricing. Also, the Carr–Madan approach is applied: it constructs a damped payoff transform $\psi(v)$, multiplies by Simpson integration weights, and executes a single FFT call to recover call prices $C(K)$ on a logarithmic strike grid. Finally, the desired call prices at arbitrary strikes are obtained and European puts $P(K)$ are derived via put–call parity $P = C - S_0 e^{-qT} + K e^{-rT}$. Figure 6 illustrates the comparison of vanilla option prices: market vs model. The left panel plots market-implied vanilla option prices (blue circles), obtained by feeding observed IVs into Black–Scholes, alongside our Bates-FFT model prices (red stars). This close alignment across strikes demonstrates that our calibrated Bates parameters successfully reproduce the level and curvature of the entire vanilla price surface.

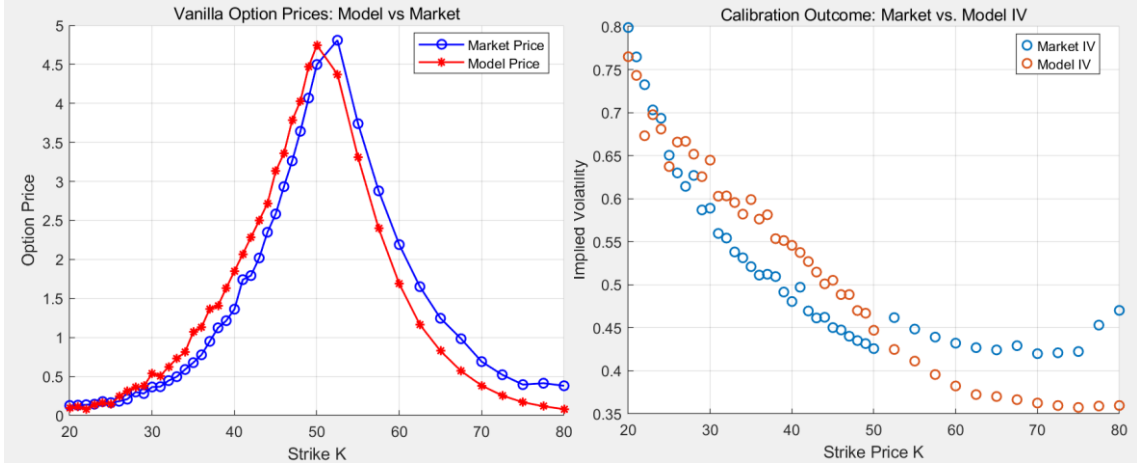


Figure 6 Vanilla Option Prices: Market vs Model

Figure 7 Calibration Outcome: Market vs Model

4.2. Bates Model Calibration on Market Vanilla Options

First, sequential calibration is used to select starting value. We calibrate the full Bates model to market-quoted vanilla option prices by solving a nonlinear least-squares problem in two sequential stages. First, we “lock down” the jump component and fit just the Heston sub-model to the cross-section of implied volatilities $\sigma_{mkt}(K_j)$, minimizing:

$$\min_{\theta} \sum_{j=1}^n w_j (\sigma_{model}(K_j; \theta) - \sigma_{mkt}(K_j))^2$$

using a trust-region Gauss–Newton solver (*lsqnonlin*) with literature-informed initials. Having locked in the continuous part, we “unlock” the jumps $(\lambda_j, \mu_j, \sigma_j)$, and rerun the same optimizer on the full eight-dimensional vector. At every iteration model prices are generated by Carr–Madan FFT, inverted to implied vols and assembled into the residual vector $r_j = \sigma_{model}(K_j; \theta) - \sigma_{mkt}(K_j)$. We terminate when the sum-of-squares reduction or parameter steps drop below tolerance (or reach max iterations). The result is an economically consistent Bates parameter vector that aligns model and market vanilla prices. Figure 7 illustrates the calibration outcomes. The right panel shows the cross-section of implied volatilities: market quotes in blue and model-inferred IVs in orange, both

plotted against strike. The market exhibits the familiar skew—high vols at low strikes that decline as K rises. Our model captures that overall shape well, slightly overshooting market vols on the downside (low strikes) and undershooting on the upside (high strikes). The residual pattern suggests there is room to fine-tune jump intensity or vol-of-vol, but overall the calibration reproduces the key features of the observed smile with high fidelity.

4.3. Monte Carlo Pricer for the Exotic Barrier Component

The Monte Carlo pricer is built on the classic Euler-Maruyama discretization of the (risk-neutral) asset SDE. Concretely, over each small step Δt we update

$$S_{t+\Delta t} = S_t \exp\left(\left(r - q - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}Z\right).$$

Each Monte Carlo script simulates the Bates jump-diffusion via Euler-Maruyama over N time steps and M paths, applying the same jump-drift-compensated drift and monitoring a down-and-out barrier H , in which payoffs are $\max(K - S_T, 0)$, if the barrier is never breached and zero otherwise. All simulated payoffs are then discounted at the risk-free rate and averaged. The plain European put at strike K is recomputed via FFT plus put-call parity, and the barrier put is priced by Monte Carlo, allowing us to report both the absolute barrier-put price and its survival probability relative to the vanilla put.

4.4. Results

After running the calibration and pricing scripts, the following key outputs are obtained:

- **Plain European Put Price**

Using Carr-Madan FFT plus put-call parity at strike $K = 1,20S_0$, we find $P \approx 10,93$.

- **Down-and-Out Barrier Put Price**

Monte Carlo simulation of the barrier put with $H = 0,8S_0$ delivers $P_{barrier} \approx 3,11$.

- **Survival Probability**

The probability that the barrier is never hit is 56.39%, which is calculated as the ratio $\frac{P_{barrier}}{P_{plain}}$,

reflecting the chance, under our calibrated Bates dynamics, that the underlying never breaches the barrier before maturity while finishing in-the-money.

4.5. Bank's Margin

As mentioned before, the bank's margin depends on how high the price is. If the price of BC equals to the sum of stock price at time t and Down-and-Out Barrier Put($K = B, H$), the net cost is certainly zero, leaving bank does not gain nor lose, in which case, the total hedging cost is $S_0 + P_{KO} = 50,46 + 3,11 = 53,57$. The issuance price and profit are determined by:

$$V_0 = (S_0 + P_{KO})(1 + M), Profit = (V_0 - S_0 - P_{KO})e^{rT}$$

Where:

M : Target profit margin (e.g., 2%)

For example, if the bank's target profit margin is 2% of hedging cost, the issuance price is $53,57 \times (1 + 2\%) = 54,64$. The amount deposited in risk-free account is $54,64 - 53,57 = 1,07$. and the bank profit, which is also the risk-free account value at T is $1,07e^{4.355\% \times 0.345} = 1,09$.

5. Hedging

5.1. Code Implements Delta-Hedging under the Bates Model

The delta-hedging strategy requires the key numerical components provided by the Monte Carlo function. The goal is to hedge the short position of the bank after issuing a Bonus Certificate (BC), whose price dynamics follow the Bates model.

We first price the BC using two parts mentioned before: underlying value and a down-and-out put component (representing the bonus barrier feature). The BC price is thus computed as:

$$price_{BC} = S_0 + price_{barrier_{put}}$$

To obtain the Delta of the BC, the code calculates a central difference approximation using a small epsilon. Specifically, it bumps the stock price up and down slightly and recomputes the BC price for both values. The Delta is then calculated as:

$$delta_{BC} = (V_{plus} - V_{minus}) / (2 * epsilon)$$

where V_{plus} and V_{minus} represent the BC prices at perturbed values of $(S_0 + epsilon)$ and $(S_0 - epsilon)$, respectively. The number of shares to buy is then computed as $Delta \times Notional$. The cash required for hedging is calculated as:

$$cashflow = cash\ from\ BC - cost\ of\ shares$$

This structure ensures that the Delta-hedge is consistent with the Bates model dynamics, accounting for stochastic volatility and jumps.

The results are, in order to implement this delta-hedging strategy, the $Delta = 0.8783$, the share needs to buy is 878.273, and the cash needed is \$2.820.386,05. A BC-delta of 0.8783 means that for each dollar notional shorted in the certificate, 0.8783 shares of the underlying stock needed to buy, in order to remain locally hedged. Scaling that to a \$1 000 000 issue, the bank must go long 878 273 shares at today's spot.

6. Conclusion

In this project we have demonstrated a full valuation and risk-management framework for a Bonus Certificate under the Bates stochastic-volatility-with-jumps model: we decomposed the product into vanilla calls/puts (priced via a Carr–Madan FFT pricer) and a down-and-out barrier put (valued by Euler–Maruyama Monte Carlo). We then calibrated all model parameters sequentially to market-observed implied volatilities via a least-squares Gauss–Newton routine. Finally, we computed the exotic payoff, extracted the certificate's delta (≈ 0.8783), and showed that a hedge consisting of 878 273 shares perfectly replicates the issuer's short position, delivering both investor guarantees and bank margin.

7. Reference

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